RESEARCH PLAN - V4.0

How Much Data Is Enough?

Abstention-Aware Menstrual Cycle Inference from Wearable Sensors

A Conformal Prediction Framework with Reject Option

Dataset: mcPHASES v1.0.0 (PhysioNet) • n = 42 (longitudinal n = 20)

Target venues: JMIR mHealth and uHealth | npj Women's Health | J Biomedical Informatics

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1. Publication Strategy

1.1 Target Abstract (Draft Zero)

This abstract defines the paper's story. If it cannot be written before coding starts, the research question is not sharp enough.

EN: Background: Wearable-based menstrual cycle inference typically assumes that sufficient data has been collected before generating predictions. In practice, data completeness and signal quality vary across users, and fixed minimum-data rules (e.g., 5 or 7 valid nights) ignore individual variation. No prior work has addressed when a wearable system has accumulated enough evidence for a specific individual to make a reliable hormone-anchored cycle phase prediction. Objective: We introduce an evidence sufficiency framework that uses conformal prediction set sizes as a calibrated uncertainty signal with an explicit reject option. The system outputs a three-valued decision: predict, defer, or request more data. Methods: Using the mcPHASES dataset (n = 42, multimodal wearable + urinary hormone data, 20 with longitudinal repeat), we compare fixed-rule baselines (3/5/7 nights), an oracle global threshold, a signal-quality heuristic, and a covariate-conditional sufficiency model. The primary endpoint is binary hormone-anchored phase (pre-ovulatory vs. post-ovulatory). We evaluate via nested leave-one-subject-out cross-validation, reporting empirical coverage rather than invoking formal conformal guarantees, which require exchangeability assumptions only approximately satisfied in this temporal setting. Expected results: The sufficiency threshold varies substantially across individuals. Covariate-conditional estimation reduces median required nights relative to the best fixed rule while maintaining controlled error. Signal completeness, cycle regularity, and modality availability are leading predictors of sufficiency speed. Conclusions: Replacing fixed data-window rules with an uncertainty-aware sufficiency framework yields safer inference for wearable menstrual health applications, with implications for product onboarding, user communication, and regulatory defensibility.

PL: **Wprowadzenie**: Wnioskowanie o fazie cyklu menstruacyjnego na podstawie danych z urządzeń wearable zazwyczaj zakłada, że przed wygenerowaniem predykcji zgromadzono wystarczającą ilość danych. W praktyce kompletność danych i jakość sygnału różnią się między użytkowniczkami, a stałe reguły minimalnej liczby obserwacji, takie jak 5 lub 7 poprawnych nocy, nie uwzględniają zmienności międzyosobniczej. Dotychczas nie opisano, w którym momencie system wearable zgromadził wystarczającą ilość dowodów, aby dla konkretnej osoby sformułować wiarygodną, predykcję fazy cyklu.

Cel: Proponujemy metodykę oceny wystarczalności dowodów, która wykorzystuje rozmiar zbiorów predykcyjnych metody predykcji konformalnej jako skalibrowany sygnał niepewności oraz uwzględnia jawną opcję wstrzymania się od predykcji. System generuje trójwartościową decyzję: przewiduj, odrocz lub zażądaj większej ilości danych.

Metody: W analizie wykorzystano zbiór mcPHASES (*n* = 42), zawierający multimodalne dane z urządzeń wearable oraz dane dotyczące stężeń hormonów oznaczanych w moczu; dla 20 uczestniczek dostępne były dodatkowo powtórzone obserwacje podłużne. Porównano strategie referencyjne oparte na stałych regułach liczby nocy (3/5/7),

pojedynczy optymalny próg globalny, heurystykę jakości sygnału oraz model wystarczalności warunkowany współzmiennymi. Główny punkt końcowy stanowiła binarna faza cyklu wyznaczona na podstawie danych hormonalnych (przedowulacyjna vs poowulacyjna). Ewaluację przeprowadzono z użyciem zagnieżdżonej walidacji krzyżowej leave-one-subject-out. Raportowano empiryczne pokrycie zamiast odwoływać się do formalnych gwarancji conformal prediction, ponieważ wymagają one założenia wymienności obserwacji, które w tym układzie czasowym jest spełnione jedynie w przybliżeniu.

Oczekiwane wyniki: Próg wystarczalności istotnie różni się między osobami. Estymacja warunkowana współzmiennymi zmniejsza medianę wymaganej liczby nocy względem najlepszej reguły stałej przy jednoczesnym utrzymaniu kontrolowanego błędu. Do najważniejszych predyktorów szybkości osiągania wystarczalności należą kompletność sygnału, regularność cyklu oraz dostępność poszczególnych modalności.

Wnioski: Zastąpienie stałych reguł opartych na długości okna danych metodyką uwzględniającą niepewność prowadzi do bezpieczniejszego wnioskowania w zastosowaniach urządzeń wearable w zdrowiu menstruacyjnym, z potencjalnymi implikacjami dla onboardingu produktu, komunikacji z użytkowniczką oraz uzasadnienia regulacyjnego.

1.2 Central Claim and Supporting Claims

EN: An uncertainty-aware framework with abstention improves menstrual phase inference compared with fixed-window rules.

PL: **Model uwzględniający niepewność i możliwość wstrzymania predykcji poprawia jakość wnioskowania o fazie cyklu menstruacyjnego względem podejść opartych na stałej liczbie obserwacji.**

Supporting claim 1

The amount of data needed for reliable inference varies across users and depends on measurable signal characteristics.

Secondary supporting claim

A sufficiency model learned from one observation period predicts convergence speed in a later period (longitudinal transfer, n = 20).

Boundary of claims (żeby nie przesadzić z obietnicą niewiadomo czego)

- This is a methodological proof-of-concept, not a clinical validation.
- All claims are conditional on a single dataset (mcPHASES, n = 42).
- We do not claim formal conformal coverage guarantees; we report empirical coverage.
- We do not claim individual-level personalization; we demonstrate covariate-conditional sufficiency estimation.
- This is not contraception guidance and must not be interpreted as such.

1.3 Pre-Specified Success and Failure Criteria

Primary success criterion: The covariate-conditional model achieves significantly higher correct-prediction rate at the same or lower median number of required nights, compared to the best fixed rule AND the oracle global threshold. Significance assessed via paired subject-level bootstrap ($p < 0.05$, 10,000 iterations).

Secondary success criterion: Even if the covariate-conditional model does not beat the oracle global threshold, demonstrating that the sufficiency threshold varies meaningfully across individuals (IQR of $\tau_i > 2$ nights) is a publishable finding about evidence heterogeneity.

If the result is negative: If no adaptive strategy outperforms Fixed-5, the paper becomes a negative-result contribution: “At the resolution of mcPHASES (n = 42), fixed rules are sufficient.” This is publishable with different framing, emphasizing conditions under which personalization does not help.

1.4 Hero Figure

Figure 1: X-axis: cumulative valid nights (1…k). Y-axis: conformal prediction set size. Individual curves per participant, color-coded by convergence speed (fast/medium/slow). Horizontal line at $|C| = 1$ (sufficiency). Vertical dashed lines at k = 3, 5, 7 (fixed rules). Shows: (a) thresholds vary, (b) fixed rules suboptimal for many, (c) some users converge fast, others slow.

This figure must be hand-sketched before any modeling begins.

1.5 Target Venues

Venue	IF	Fit
JMIR mHealth and uHealth	~5	Best fit. Wearable + ML + proof-of-concept accepted. Thigpen et al. 2025 published here.
npj Women’s Health	New	Kilungeja et al. 2025 used same dataset. Must clearly differentiate question.
J Biomedical Informatics	~4.5	Good if framing leans methodological (UQ framework).
ML4H / ACM CHIL	Workshop	Backup: shorter format, fast review, ML-for-health audience.

2. Scientific Framing

2.1 Research Question

> *How much wearable data is sufficient for stable hormone-anchored menstrual cycle phase inference, and does the sufficiency threshold vary across individuals in ways that can be predicted from observable signal characteristics?*

2.2 Literature Gap Statement

This paragraph must be writable before WP5 begins. It is the paper's novelty anchor.

> *Existing work on menstrual phase classification from wearable data evaluates predictive accuracy assuming data completeness or after excluding cycles with missing data (Goodale et al. 2019; Kilungeja et al. 2025; Masuda et al. 2025; Thigpen et al. 2025). None addresses a prior question: how many observation nights are sufficient for a specific individual before a prediction should be made at all? This omission matters because (a) wearable health products must decide during onboarding when to deliver a first prediction, (b) real-world adherence varies across users, and (c) regulatory frameworks for software as a medical device increasingly require uncertainty quantification under data limitations (EU MDR 2017/745). In sleep research, Lau et al. (2022) showed that the number of actigraphy nights needed for reliable estimates varies by metric, establishing a minimum-data precedent that has not been transferred to menstrual inference. In machine learning, selective prediction (Swaminathan et al. 2024) and conformal prediction (Romano et al. 2020) provide formal tools for abstention-aware classification. Punzi and Thuis (2025) have raised ethical concerns about algorithmic overconfidence in fertility tracking, motivating mechanisms that allow the system to say "I don't know yet." This study bridges these strands.*

2.3 Formal Definition of Sufficiency

Primary definition: For participant i, after k valid nights, compute the conformal prediction set $C_\alpha(x_i^{1:k})$ at nominal level $1 - \alpha = 0.90$. Define the sufficiency point τ_i as the smallest k such that $|C_\alpha| = 1$ for at least 2 consecutive observation windows. This definition is operational, not a formal guarantee - we validate empirical coverage via LOSO-CV rather than invoking theoretical exchangeability.

Decision policy: $|C_\alpha| = 1 \rightarrow$ predict. $|C_\alpha| > 1 \rightarrow$ defer. Sensor dropout or quality flags → request more nights.

Sensitivity definition: Replace conformal set size with calibrated posterior entropy: $H(p) < 0.5$ bits for 2 consecutive windows. Report both; conformal set size is primary.

2.4 Key Methodological Honesty: Conformal Prediction in a Temporal Setting

This subsection must appear in the Methods and be referenced in Limitations. It is the single biggest reviewer risk.

> *Standard split conformal prediction provides distribution-free coverage guarantees under exchangeability of calibration and test data (Vovk et al. 2005). In our setting, this assumption is only approximately satisfied: observations accumulate sequentially within a cycle, consecutive nights are autocorrelated, and the test scenario (partial cycle from a new user) differs structurally from the calibration scenario (complete cycles from other users). Stocker et al. (2025) explicitly caution against naïve application of conformal methods to temporally dependent data. Accordingly, we do NOT claim formal coverage guarantees. Instead, we use the conformal prediction set size as a calibrated uncertainty signal and validate empirical coverage via leave-one-subject-out cross-validation. The validity of our approach rests on empirical evidence, not on the theoretical guarantee.*

2.5 Differentiation from Kilungeja et al. (2025) - to jest niebezpieczny paper - bo “w pobliżu”

This is critical because Kilungeja et al. used the same dataset (mcPHASES) and the same wearable modalities.

Dimension	Kilungeja et al. 2025	This Study
Question	How accurately can we classify menstrual phases?	How much data is needed before we should classify at all?
Output	Phase label (always produced)	Three-valued decision: predict / defer / collect more
Endpoint	3-phase and 4-phase classification accuracy	Binary pre-/post-ovulatory with sufficiency threshold
Uncertainty	Not modeled	Conformal prediction set size as uncertainty signal
Abstention	Not available	Explicit reject option when data insufficient
Evaluation	Accuracy, AUC-ROC	Coverage-error tradeoff, median τ, calibration, deferral rate
Novelty type	Classification model	Decision-theoretic framework

3. Methodological Specification

3.1 Primary Endpoint

Primary: Binary phase label - pre-ovulatory vs. post-ovulatory - anchored to urinary LH surge from Mira Plus data. The post-ovulatory state is confirmed by subsequent PdG rise. This binary endpoint maximizes signal-to-noise (the temperature/HR shift after ovulation is the strongest physiological signal) and is more clinically relevant (fertile window determination).

Secondary (supplement): (a) 4-phase (menstrual, follicular, ovulatory, luteal). (b) 3-phase (menstrual, follicular+ovulatory, luteal). (c) Binary derived from temperature biphasic pattern only (no hormones). All three reported as robustness checks.

3.2 Base Classifier

Calibrated gradient-boosted ensemble (LightGBM) trained on nightly feature vectors. Features: nocturnal skin temperature statistics, HR/HRV summary metrics, EDA tonic level, continuous glucose summary stats, self-reported symptom indicators. Probability calibration via isotonic regression on a held-out inner fold.

This is not the contribution. It is infrastructure. Student 2 builds the base classifier; Student 3 builds the sufficiency layer on top.

3.3 Uncertainty Quantification Layer

We apply split conformal prediction with Adaptive Prediction Sets (Romano et al. 2020) as the uncertainty layer. For each participant i and cumulative window of k valid nights:

1. Compute non-conformity scores $s(x, y) = 1 - \hat{p}(y \mid x)$ on the calibration fold.
2. Determine conformal quantile $\hat{q}$ at level $\lceil (n+1)(1-\alpha) \rceil / n$.
3. Construct prediction set $C_\alpha(x_{test}) = \{y : \hat{p}(y \mid x_{test}) \geq 1 - \hat{q}\}$.
4. Record $|C_\alpha|$ as the sufficiency signal. $|C_\alpha| = 1$ means high confidence; $|C_\alpha| = 2$ means both classes plausible.

Critical framing: We use $|C_\alpha|$ as a calibrated uncertainty signal, not as a formal probabilistic guarantee. Empirical coverage is validated via LOSO-CV and reported as a primary result.

3.4 Covariate-Conditional Sufficiency Estimation

This is the core contribution. For each user, we model how quickly the conformal set shrinks to a singleton as a function of covariates:

- Signal completeness: proportion of valid nights in the accumulation window.
- Cycle regularity: CV of prior cycle lengths (if available from Round 1).
- Modality availability: binary flags for each sensor stream present.
- Nocturnal signal quality: mean SNR of temperature and HR signals.
- Prior history: features from Round 1 for participants with longitudinal data.

Approach: Mondrian conformal prediction (stratified by covariate groups) as the primary method. This is transparent, avoids overfitting at n = 42, and allows interpretable reporting of within-stratum sample sizes. As a sensitivity analysis, also fit a simple regression meta-model predicting τ_i from covariates.

Why not "personalization": At n = 42, true individual-level adaptation is not defensible. Covariate-conditional estimation adjusts the sufficiency threshold based on observable characteristics - which is what a real product would do during onboarding. This framing is honest and practical.

3.5 Comparators (Expanded)

Strategy	Description	Role
Fixed-3	Predict after 3 valid nights. No deferral.	Lower bound
Fixed-5	Predict after 5 valid nights. No deferral.	Primary fixed baseline
Fixed-7	Predict after 7 valid nights. No deferral.	Upper bound
Always-predict	Predict at k = 1. Never defer.	Shows cost of never abstaining
Oracle global threshold	Single learned τ^* minimizing expected loss over training set.	Critical: shows whether ANY adaptivity beats the best fixed k
Signal-quality heuristic	Predict when mean nocturnal temp variance < threshold. Simple, no ML.	Shows whether conformal machinery is needed at all
Covariate-conditional	Sufficiency threshold adjusted by observable covariates.	Proposed method

The oracle global threshold and signal-quality heuristic are essential. Without them, a reviewer will argue that the covariate-conditional model is simply finding a better universal threshold rather than truly conditioning on individual characteristics.

3.6 Validation Strategy

Primary: Nested leave-one-subject-out cross-validation (nested LOSO-CV). Outer loop: 42 iterations. Inner loop: LOSO on remaining 41 for hyperparameters. This is the only credible strategy at n = 42.

Longitudinal: For 20 participants with Round 2 data: train on Round 1, evaluate on Round 2. Tests temporal transfer.

Confidence intervals: Subject-level bootstrap, 10,000 iterations, BCa-corrected.

Computational feasibility check: Estimate walltime for a single LOSO fold in week 4. If total nested LOSO > 24 hours, fall back to 5-fold grouped CV with subject-level grouping.

3.7 Evaluation Metrics

Metric	Purpose	Priority
Median τ (required nights)	Efficiency: how quickly does the system reach sufficiency?	Primary

Metric	Purpose	Priority
Empirical coverage	Validity: does the conformal set contain the true label ≥90% of the time?	Primary
Selective accuracy	Accuracy among predicted (non-deferred) instances.	Primary
Coverage-error tradeoff curve	Full tradeoff between prediction rate and error.	Primary (Fig 2)
IQR of τ_i across participants	Evidence heterogeneity: does sufficiency actually vary?	Primary
Calibration (reliability diagram)	Is estimated uncertainty well-calibrated?	Secondary (Fig 3)
Deferral rate	Proportion of windows where the system abstains.	Secondary
Covariate associations with τ_i	Which characteristics predict faster/slower convergence?	Secondary

4. Data Plan: mcPHASES

4.1 Dataset Overview

mcPHASES (Lin et al. 2026) is a publicly available multimodal dataset on PhysioNet. 42 Canadian young adults, 3-month observation (Round 1), 20 with a second 3-month round (Round 2). 23 structured tables. Fitbit Sense (temperature, HR, HRV, EDA, sleep), Dexcom G6 CGM (glucose), Mira Plus urinalysis (LH, E3G, PdG), daily surveys (symptoms).

4.2 Endpoint Construction (Critical Path)

This is the single most important technical decision. If the endpoint is weak, the paper is rejected.

1. Identify the LH surge day from raw Mira urinalysis data for each cycle.
2. Confirm post-ovulatory state via subsequent PdG rise (≥2 days of elevated PdG).
3. Assign binary label: pre-ovulatory (all days before LH surge) vs. post-ovulatory (LH surge day onward through end of luteal phase).
4. Quantify label confidence: for what proportion of cycle-days is the assignment unambiguous?
5. If >30% of cycle-days have ambiguous labels, escalate to supervisor and consider switching to a simpler endpoint.

Critical rule: Do NOT use derived labels from mcPHASES without independent verification against raw hormone data. Student 1 must reconstruct labels from raw LH/E3G/PdG values and compare with any provided labels.

4.3 Valid Night Definition

Proposed: ≥4 hours of continuous Fitbit recording during 22:00-06:00. This must be frozen in WP2 and applied consistently throughout.

5. Work Packages

WP1. Protocol Freeze

Owner: Supervisor + full team. Weeks 0-2. Includes synthetic data generator design (day 2-3).

Freeze: research question, claims, endpoint, validation strategy, success criteria, hero figure design. Exit criterion: every team member can state the central claim and primary metric from memory.

Deliverables

1. Finalized protocol (this document, signed off).
2. Hand-drawn hero figure sketch with simulated data.
3. Gap statement paragraph (Section 2.2) finalized.
4. Claims boundary document.
5. All team members complete Angelopoulos & Bates (2021) conformal prediction tutorial.

WP2. Data Audit, Preprocessing, Cohort Definition

Owner: Student 1. Weeks 0-4.

Scope

- Import all mcPHASES tables. Produce data dictionary.
- Quantify missingness at participant / cycle / night / modality level.
- Define valid night operationally. Apply inclusion/exclusion criteria.
- Identify Round 1 vs Round 2 participants and inter-round gap.
- Freeze LOSO folds and save to disk.
- Build reproducible preprocessing pipeline (scripts, not notebooks).

Deliverables

Clean dataset, frozen cohort, missingness report, valid-night definition, LOSO fold files, preprocessing pipeline, Table 1 (cohort characteristics).

WP3. Endpoint Construction and Ground-Truth Validation

Owner: Student 1. Weeks 2-4 (continues from WP2, parallel with synthetic pipeline work by S2-S4).

Scope

- Construct primary binary label from raw Mira LH + PdG data (Section 4.2).
- Quantify label confidence per cycle-day.
- Prepare three sensitivity endpoints (4-phase, 3-phase, temperature-only binary).
- Document endpoint protocol step-by-step for independent replication.

Critical warning

If endpoint relies on mcPHASES-derived labels without raw hormone verification, reviewers will challenge the entire paper. Student 1 must reconstruct from raw data.

WP4. Literature Review and Positioning

Owner: All students (shared task). Weeks 1-4. Each student reviews 3-4 papers from their technical domain.

Required literature matrix dimensions

- Data type • Ground-truth standard • Model type • Uncertainty modeling • Abstention capability • Missing data strategy

Required citations by role

Role	Papers
Direct comparators (S2 reviews)	Kilungeja et al. 2025, Masuda et al. 2025, Thigpen et al. 2025
Background (S1 reviews dataset/endpoint papers; S3 reviews methods)	Goodale et al. 2019, Lyzwinski et al. 2024, Shi et al. 2026, Wang et al. 2025
Abstention/sufficiency framing (S3 reviews)	Swaminathan et al. 2024, Punzi & Thuis 2025, Lau et al. 2022
Methods references (S3 reviews)	Vovk et al. 2005, Romano et al. 2020, Stocker et al. 2025
Dataset (S1 reviews)	Lin et al. 2026

Deliverables

Literature matrix table (assembled from all students' contributions), draft Introduction with gap statement, Related Work section, novelty statement. Each student delivers 3-4 paper summaries from their domain by week 3.

WP5. Baselines and Fixed-Rule Analysis

Owner: Student 2. Weeks 0-6 (synthetic pipeline weeks 0-3, real data weeks 4-6).

Implementation list

1. Train base classifier (LightGBM) with isotonic calibration under nested LOSO-CV.
2. Implement Fixed-3, Fixed-5, Fixed-7 baselines (predict after k nights, no deferral).
3. Implement Always-predict baseline (predict at k = 1).
4. Implement Oracle global threshold: find single τ^* minimizing loss on training set.
5. Implement Signal-quality heuristic: predict when mean nocturnal temperature variance < learned threshold.
6. Compute all metrics (Section 3.7) for every baseline.
7. Produce first benchmark table and performance-vs-nights curves.

Constraint

Exactly 6 baselines. No more. No "model zoo."

Critical checkpoint: computational feasibility

After the first LOSO fold completes, estimate total walltime for nested LOSO-CV. If > 24 hours, notify supervisor and switch to 5-fold grouped CV.

WP6. Covariate-Conditional Sufficiency Model (Core Contribution)

Owner: Student 3. Weeks 0-8 (synthetic conformal layer weeks 0-4, real model weeks 5-8).

Implementation steps

1. Apply split conformal prediction with APS on top of the calibrated base classifier from WP5.
2. For each participant and each k (cumulative nights), compute $C_\alpha(x_i^{1:k})$ and record $|C_\alpha|$.
3. Produce individual sufficiency curves ($|C_\alpha|$ vs k). This is the raw material for Figure 1.
4. Identify τ_i for each participant (first k where $|C_\alpha| = 1$ for 2 consecutive windows).
5. Implement Mondrian conformal prediction: stratify participants by covariate groups (e.g., high/low signal completeness, regular/irregular cycles). Report within-stratum sample sizes.
6. As sensitivity: fit a simple regression predicting τ_i from covariates.
7. Implement full decision policy: predict / defer / collect more.
8. Compare against ALL baselines from WP5 under nested LOSO-CV.
9. Produce hero figure (Figure 1) with real data.

Deliverables

Sufficiency algorithm, decision policy implementation, hero figure, main comparison table, covariate association analysis.

What Student 3 must understand

> *Your job is not to build the best classifier. Your job is to build the system that knows when it does not know. Student 2 delivers the calibrated base classifier. Your contribution is the conformal sufficiency layer and decision policy on top.*

WP7. Statistical Evaluation, Calibration, and Robustness

Owner: Student 4. Weeks 0-10 (evaluation harness on synthetic outputs weeks 0-5, real analyses weeks 6-10).

Required analyses

1. All primary metrics with bootstrap CIs (subject-level, 10,000 iterations, BCa).
2. Empirical coverage analysis: does the conformal set contain the true label ≥90% of the time? Report per-stratum and overall.
3. Calibration: reliability diagrams for the base classifier.
4. Coverage-error tradeoff curves for all strategies (Figure 2).

5. Sensitivity to endpoint definition: repeat with all 3 secondary endpoints.
6. Sensitivity to missingness: artificially increase dropout (10%, 20%, 30%) and re-evaluate.
7. Modality ablation: remove temperature / HR-HRV / glucose / EDA / self-report one at a time. Measure impact on median τ.
8. Longitudinal transfer: train on Round 1, evaluate on Round 2 (n = 20). Figure 4.
9. Subgroup analysis: regular vs. irregular cycles (if subgroups large enough).

Critical note

> *A reviewer may accept moderate model performance. A reviewer will NOT accept poor calibration, uncontrolled uncertainty, or overclaimed results.* ***Your job is to make the paper's claims defensible, not to make the numbers look good.***

WP8. Paper Writing, Figures, Supplement, Reproducibility (Shared)

Owner: All students. Weeks 8-12. Each writes Methods/Results for their own WP. Supervisor integrates and edits.

Paper budget

4,000-5,000 words. 4 figures. 2 tables. Everything else in supplement.

Element	Content	Owner
Figure 1	Hero: individual sufficiency curves with fixed-rule markers	Student 3
Figure 2	Coverage-error tradeoff for all 7 strategies	Student 4
Figure 3	Calibration reliability diagrams	Student 4
Figure 4	Longitudinal transfer: Round 1 → Round 2	Student 4
Table 1	Cohort characteristics	Student 1
Table 2	Main comparison: all strategies, all primary metrics with CIs	Students 2+3+4

Limitations (draft now, before results)

- n = 42 (longitudinal n = 20). Proof-of-concept, not clinical validation.
- Single dataset, single device (Fitbit Sense), single population (young Canadian adults).
- Conformal exchangeability approximately satisfied; empirical coverage reported, not formal guarantees.
- Mira Plus urinalysis is not blood-draw quality; hormone ground truth has uncertainty.
- Adherence gaps likely not missing at random.
- Does not address hormonal contraception, PCOS, endometriosis, or perimenopause.
- No external validation cohort.

6. Team Structure (4 Students)

Role	Primary WP	Key Deliverables
Student 1	WP2 + WP3 (Data + Endpoint)	Clean dataset, cohort, LOSO folds, Table 1, binary labels from raw hormones, sensitivity endpoints
Student 2	WP5 (Baselines + Base Classifier)	Base classifier, 6 baseline strategies, benchmark table, LOSO-CV pipeline
Student 3	WP6 (Conformal Layer + Sufficiency)	Conformal prediction layer, sufficiency curves, hero figure, decision policy, covariate analysis
Student 4	WP7 (Statistics + Robustness)	Bootstrap CIs, calibration, coverage-error tradeoff, longitudinal transfer, ablation, all evaluation figures

All four students do technical work. Each writes the Methods and Results text for their own WP. Literature review is shared: each student reviews 3-4 papers from their domain. Supervisor integrates the manuscript draft and reviews all claims for proportionality.

7. Timeline (12 Weeks)

Week	Active WPs	Milestone
1-2	WP1 + WP2 + synthetic pipelines	Protocol frozen. Synthetic data generator built (S1+S2, day 2-3). Hero figure sketched. Conformal tutorial studied by all. S2-S4 begin building pipelines on synthetic data.
2-4	WP2 + WP3 + WP5/6/7 (synthetic)	S1: Clean dataset delivered, binary endpoint, LOSO folds frozen. S2: ML pipeline running on synthetic data. S3: Conformal layer running on synthetic data. S4: Evaluation harness tested on fake outputs.
3-5	WP4 (shared) + WP5 (real data)	Literature reviews delivered by all (3-4 papers each). Gap statement written. S2: base classifier trained on real data. S3: conformal layer ready to receive real classifier.
4-6	WP5	S2: all 6 baselines evaluated on real data. First benchmark table. Computational feasibility check for nested LOSO-CV.
5-8	WP6	S3: Conformal layer applied to real classifier. Sufficiency curves computed. Hero figure with real data. Decision policy implemented.
6-10	WP7	S4: Robustness, calibration, longitudinal transfer, ablation. All analyses complete by week 10.
8-12	WP8	Each student writes their section. Supervisor integrates. Internal review week 11. Final version week 12.

8. Mandatory Analyses

ID	Analysis	Location	Owner
A	Data completeness and adherence	Supplement	S1
B	Fixed-rule performance (3/5/7 + always-predict)	Main (Table 2)	S2
C	Oracle global threshold	Main (Table 2)	S2
D	Signal-quality heuristic	Main (Table 2)	S2
E	Covariate-conditional sufficiency model	Main (Table 2, Fig 1)	S3
F	Coverage-error tradeoff	Main (Figure 2)	S4
G	Empirical conformal coverage	Main (text + Table 2)	S4
H	Calibration	Main (Figure 3)	S4
I	Longitudinal transfer (R1→R2)	Main (Figure 4)	S4
J	Sensitivity: endpoint definitions	Supplement	S1+S4
K	Modality ablation	Supplement	S4
L	Sensitivity: artificial missingness	Supplement	S4
M	Covariate associations with τ_i	Main or supplement	S3

Analyses A-I are required for the main paper. J-M are required for the supplement.

9. Team Rules

9.1 Data integrity

1. LOSO folds generated once (WP2), saved to disk. No experiment uses different splits.
2. No model sees test-fold data during training or tuning. Student 4 audits.
3. Every experiment logged: date, config hash, metrics, responsible person.

9.2 Scope control

1. No new model or analysis unless it answers the central claim or a pre-specified supporting claim.
2. Research question, endpoint, validation, success criteria frozen after WP1. Changes require supervisor sign-off.

9.3 Reproducibility

1. Every result reproducible from repo with a single command. Seeds fixed.
2. Final repo: preprocessing, training, evaluation, figure-generation scripts, README.

9.4 Scientific integrity

1. Claims proportional to evidence. Proof-of-concept framing, not clinical validation.
2. Negative results reported honestly.
3. Every result ends as figure, table, or Results/Discussion paragraph. If not, it was unnecessary.

10. Risks

Risk	Severity	Mitigation
Reviewer challenges conformal guarantees	Critical	Already addressed: claim empirical coverage, not formal guarantee. Cite Stocker 2025.
Weak endpoint	Critical	WP3 high priority. If >30% ambiguous, switch to simpler endpoint.
Kilungeja et al. differentiation unclear	High	Section 2.5 differentiation table must appear in Introduction.
n = 42 limits power	High	Proof-of-concept framing. Nested LOSO. Report IQR of τ_i as evidence of heterogeneity even if comparison is underpowered.
Scope creep	High	Freeze after WP1. “Does this answer the central claim?” gate.
Data leakage	High	LOSO folds frozen in WP2. Student 4 audits.
Conformal prediction unfamiliar	Medium	Week 1-2: all study tutorial. Student 3 presents a conformal prediction walkthrough in week 2.
Nested LOSO too slow	Low	Feasibility check week 4-5. Fallback: 5-fold grouped CV.

11. Optional: Synthetic Simulation Study

Recommended but not required. Addresses the reviewer question: “Would this work at larger n?”

Design

Generate 200-1,000 synthetic users with known cycle regularity, signal quality, and phase-dependent physiological profiles. True sufficiency threshold is computable because ground truth is known.

Purpose

- Verify the method recovers true thresholds.
- Show behavior as n increases.
- Provide calibration benchmark with known ground truth.

Placement

If strong: Methods subsection + supplementary figure. If mixed: supplement only.

12. Expected Final Outputs

12.1 Main Paper

4,000-5,000 words. Abstract + Introduction + Related Work + Methods + Results + Discussion + Limitations. 4 figures, 2 tables.

12.2 Supplement

- Data completeness report
- Endpoint construction protocol
- Sensitivity: alternative endpoints
- Modality ablation
- Artificial missingness sensitivity
- Literature comparison matrix
- Simulation results (if conducted)

12.3 Reproducibility Package

Complete pipeline, frozen folds, figure scripts, README with setup and expected runtime.

13. Core Message

> *This project does not ask: “How accurately can we classify menstrual phases?” That question has been answered (Goodale 2019, Kilungeja 2025, Masuda 2025, Thigpen 2025). This project asks a harder and more important question: “When does the system have enough evidence to classify at all, and when should it wait?” That is the scientific contribution. That is what makes it publishable. That is what no prior paper has done.*

> *Before you write any code, read the abstract. Before you train any model, draw (even in the imagination) the hero figure. Before you report any result, ask: Does this answer the central claim? If not, it does not belong in this paper.*

> *Be honest about what you cannot claim. You cannot claim formal conformal guarantees in this setting. You cannot claim clinical validation at n = 42. You cannot claim individual-level personalization. What you CAN claim: a principled framework for evidence sufficiency that produces safer inference than fixed rules, with empirical evidence that sufficiency varies across users. That is enough for a good paper.*